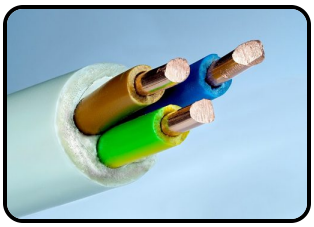
💡 Conductivity Lab: What Lets Electricity Through?

Chemistry · Unit 1C · Level 2



Name: _____

L1.13 No Swimming (salt water + light bulb) and L1.13.5 You Light Up My Life (conductivity tester with solids and solutions) · check with Mr. Brittle.

🧺 Test these

- ☐ 💧 **1 cup** plain water
- ☐ 🧂 **1 cup** salt water
- ☐ 🍬 **1 cup** sugar water
- ☐ 🔩 **1** piece of foil or copper
- ☐ 🕯️ **1** piece of wax (paraffin)

🌱 Steps: check each box when you finish

1.  Goggles on. Check the tester lights when its two ends touch. ☐
2.  **Predict**: will each thing conduct? Write yes or no. ☐
3.  **Test the solids**: foil/copper, wax, sugar, salt crystals. ☐
4.  **Test the liquids**: plain water, then salt water (1%, 5%, 10% if available). ☐
5.  **Test** sugar water. ☐
6.  **Record**: bright, dim, or off. ☐
7.  Dry the tester between every test. ☐
8.  **Sort**: conductors vs. insulators. What do the conductors have in common? ☐

📋 Data table

Material	Prediction	Bulb: bright / dim / off	Conductor?
foil / copper			
wax			
plain water			
1% salt water			
10% salt water			
sugar water			